

Parameterized ALU

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- **Project Introduction**

The project implements a parameterized Arithmetic Logic Unit (ALU) using Verilog. The ALU supports both arithmetic and logical operations based on a control signal called MODE. The ALU inputs consist of two operands (OPA and OPB), control signals (CMD, MODE, CE, RST, INP_VALID) and carry in (CIN) to generate corresponding outputs including result, carry out, overflow, and error indication.

The design also incorporates different timing behaviours:

- Most operations have 1 cycle latency
- Multiplication operations have 2 cycle processing latency (3 cycle total timing)

When MODE = 1, the ALU performs arithmetic operations selected using the CMD signal. These operations include unsigned addition, subtraction, addition with carry input, subtraction with carry, increment and decrement operations on operands, comparison operations, multiplication-related operations, signed addition, and signed subtraction. For signed arithmetic operations, the ALU also generates status flags such as overflow, negative, zero, greater-than, less-than, and equality indicators. When MODE = 0, the ALU performs logical and bit manipulation operations such as AND, NAND, OR, NOR, XOR, XNOR, NOT, logical shifts, and rotate operations. The rotate functionality is implemented in a parameterized manner where the rotation amount depends on specific lower bits of operand OPB based on the configured operand width. Invalid higher bits during rotate operations generate an error condition through the ERR output signal.

The design includes several input signals such as operand A (OPA), operand B (OPB), carry input (CIN), clock (CLK), asynchronous active-high reset (RST), clock enable (CE), operation mode (MODE), input validity control (INP_VALID), and

command selection (CMD). The outputs generated by the ALU include the operation result (RES), carry-out flag (COUT), overflow flag (OFLOW), comparison flags (G, L, E), and error indication (ERR). The INP_VALID signal is used to indicate whether operand A, operand B, or both operands are valid for the current operation.

The timing behaviour of the ALU is also an important part of the design. All operations except multiplication require one clock cycle latency, where inputs are applied during one clock cycle and the corresponding output is generated in the next clock cycle. At the same time, the ALU can accept new inputs for continuous pipelined operation.

Multiplication-related operations require additional processing time and therefore follow a multi-cycle execution. In these operations, inputs are provided during the first clock cycle, internal processing occurs during the second cycle, and the result is generated during the third clock cycle. New inputs may also be accepted along with the output generation cycle for efficient operation flow.

The reset signal is implemented as an asynchronous active-high reset. Whenever RST becomes high, all ongoing operations are cleared, and outputs are reset immediately independent of the clock. The clock enable signal CE is treated as an active-high disable control signal. When CE is high, the ALU temporarily halts operation and retains its current state until CE becomes low again. Additionally, the control signals MODE and CMD are given high priority, meaning that if these signals change before the active clock edge, the ALU performs the updated operation at the next positive edge of the clock.

The verification objective of this project is to ensure correct functionality of all arithmetic and logical operations under different operating conditions. The verification process focuses on validating signed and unsigned computations, timing behaviour, multi-cycle multiplication operations, overflow and carry generation, comparison flags, rotate error handling, reset behaviour, clock enable functionality, and parameterized operation for multiple data widths. The project follows a standard RTL design and verification methodology that includes requirement analysis, architecture development, RTL implementation, testbench creation, functional simulation, corner-case testing, waveform analysis, and debugging. Through this project, a complete understanding of ALU design, parameterized development, and functional verification flow is achieved.

- **Objectives:**

- To design and implement a parameterized ALU capable of performing multiple arithmetic and logical operations.
- To implement arithmetic operations such as addition, subtraction, increment, decrement, comparison, shifting and multiplication
- To implement logical operations including AND, NAND, OR, NOR, XOR, XNOR, NOT, shift and rotate operations.
- To perform signed and unsigned arithmetic operations.
- To generate outputs such as COUT, OFLOW, G, L, E and ERR.
- To design single and multi-cycle execution for different ALU operations.
- To implement asynchronous reset and clock-enable control in the ALU architecture.
- To verify the functionality of the ALU using Verilog testbenches.
- To validate ALU behaviour for normal, boundary and corner-case conditions.
- To analyze timing behaviour and functional correctness through simulation.
- To perform coverage analysis for complete verification of ALU operations.

- **Design Architecture**

The designed module is a parameterized Arithmetic Logic Unit (ALU) implemented using Verilog. The ALU performs both arithmetic and logical operations based on the MODE. The ALU operates synchronously with the clock signal and includes active high reset functionality. The design also supports clock enable control and input validity checking for proper operation execution.

Inputs:

SIGNAL	WIDTH	DESCRIPTION
OPA	Parameterized	Operand A input to the ALU
OPB	Parameterized	Operand B input to the ALU
CIN	1 bit	Carry input used in arithmetic operations
CLK	1 bit	System clock signal used for synchronous operation of the ALU
RST	1 bit	Asynchronous active high reset

CE	1 bit	Clock Enable Signal Processing occurs when CE is active
MODE	1 bit	MODE=0 Logical operations MODE=1 Arithmetic operations
INP_VALID	1 bit	Indicates if the inputs are valid or not
CMD	Parameterized	Commands to select operations to be performed

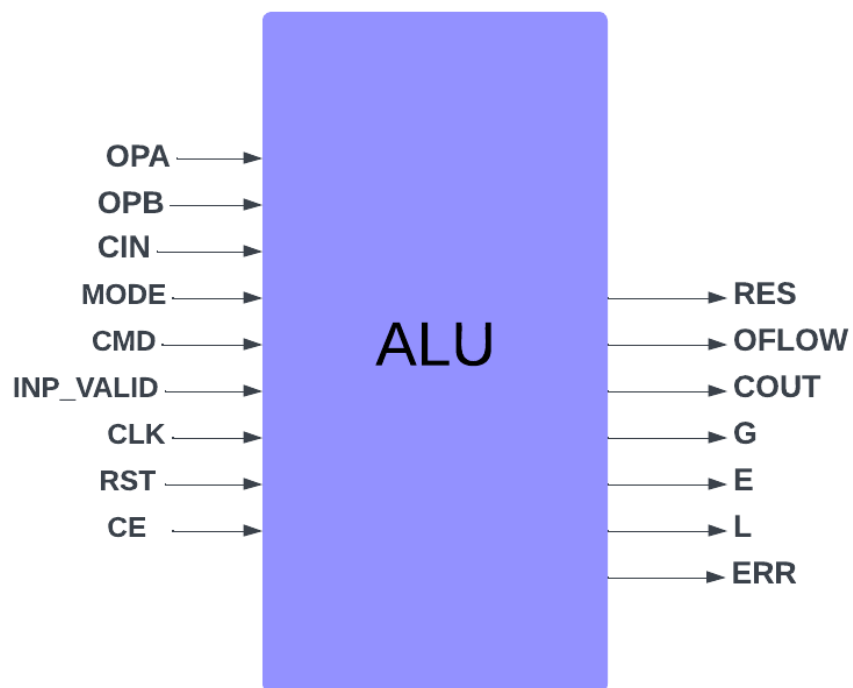
INP_VALID:

00	No valid operand
01	Only Operand A valid
10	Only Operand B valid
11	Both Operand A and B valid

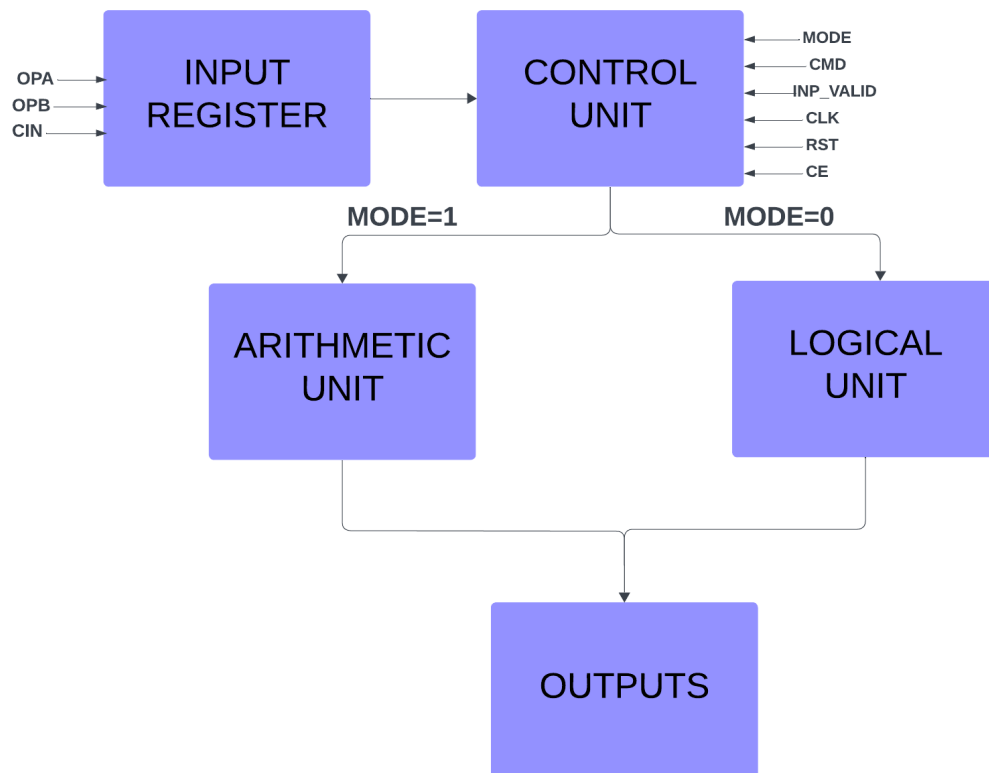
Outputs:

SIGNAL	WIDTH	DESCRIPTION
RES	Parameterized	Result generated by the ALU after performing selected operations
OFLOW	1 bit	Overflow flag generated during signed arithmetic operations
COUT	1 bit	Carry output generated during unsigned arithmetic operations
G	1 bit	Greater than flag generated during comparison operations
L	1 bit	Less than flag generated during comparison operations
E	1 bit	Equal to flag generated during comparison operations
ERR	1 bit	Error flag generated for invalid conditions

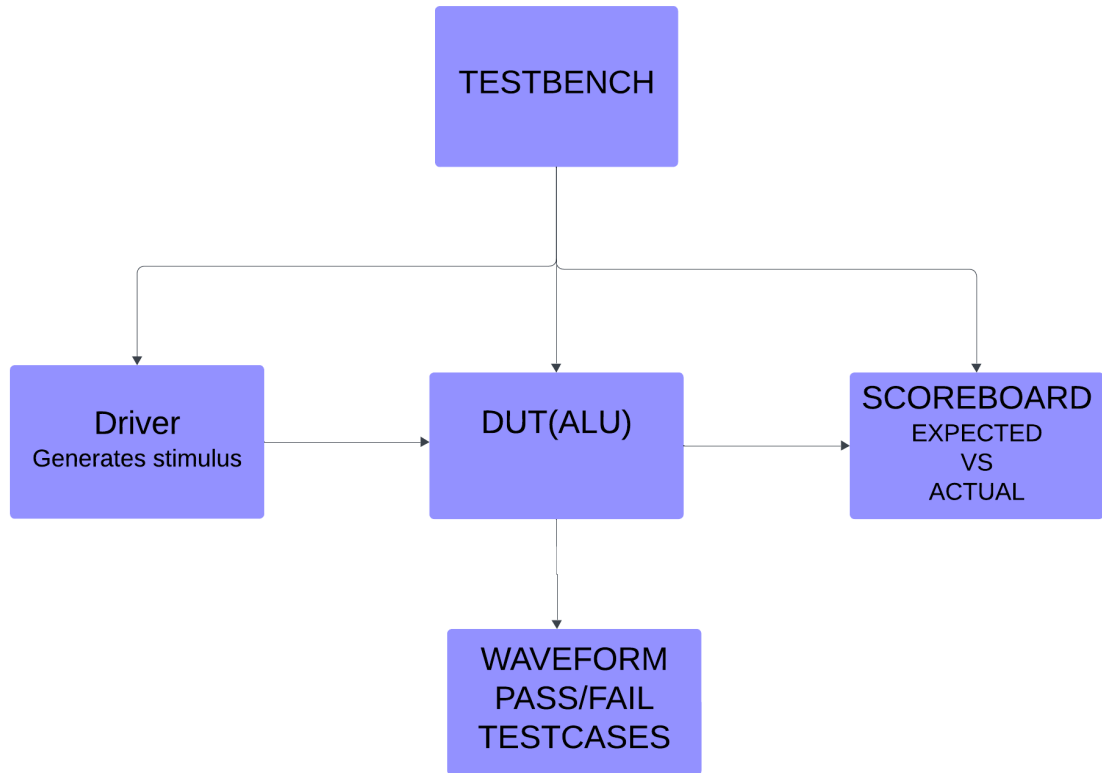
ALU BLOCK DIAGRAM



EXPANDED ALU BLOCK DIAGRAM



Testbench Architecture:



- **Timing Behaviour :**

The ALU operates synchronously with respect to the positive edge of the clock signal. The design supports both single cycle and multiple cycle operations depending on the selected commands.

Input operands are first captured into internal registers before execution.

All arithmetic and logical operations except multiplication require one clock cycle delay

- Cycle 1: Inputs and control signals are captured
- Cycle 2: Result and output flags are generated

The ALU can simultaneously accept new inputs while producing the current output.

Multiplication related operations (CMD=9 AND CMD=10) require three clock cycle for execution.

- Cycle 1: Inputs are captured
- Cycle 2: Internal shifting or increment operation
- Cycle 3: Multiplication and final result generated

During the final output cycle, the ALU can accept the next input.

Operations:

The ALU operations are divided into two major categories based on the MODE signal.

- Arithmetic Operations (MODE=1):
These operations perform mathematical operations on the input operands.

CMD	Operations
0000	Addition (Unsigned)
0001	Subtraction (Unsigned)
0010	Addition with carry in
0011	Subtraction with carry in
0100	Increment operand A
0101	Decrement operand A
0110	Increment operand B
0111	Decrement operand B
1000	Compare Operands
1001	Increment both operands and Multiply
1010	Left shift operand A and Multiply
1011	Signed Addition
1100	Signed Subtraction

- Logical Operations (MODE=0):
These operations perform bitwise operations and shift operations on the input operands.

CMD	Operations
0000	AND
0001	NAND
0010	OR
0011	NOR

0100	XOR
0101	XNOR
0110	NOT A
0111	NOT B
1000	Right Shift operand A
1001	Left Shift operand A
1010	Right Shift operand B
1011	Left Shift operand B
1100	Rotate Left A by B
1101	Rotate Right A by B

Working:

The operation of the ALU can be divided into three major phases:

- **Input Phase:**
During this phase, input operands (OPA and OPB) are applied, control signals are provided and input validity is checked using INP_VALID. If invalid inputs are detected, the ERR flag is asserted.
- **Operation Phase:**
The control unit decoded the MODE and CMD signals. Based on the selected mode arithmetic or logical operations are executed. Multiplication operations enter a multi cycle processing stage controlled by internal counter.
- **Output Phase:**
The computed result is stored in RES output register. Other output flags such as OFLOW, COUT, G, E, L and ERR are updated. The ALU accepts next inputs. For multiplication operations, outputs are generated at the positive edge of third clock cycle, while all other operations generate outputs after one clock cycle delay.
- **Testbench Components:**
The verification environment for the parameterized ALU is divided into multiple components such as top module, driver and scoreboard. Each

component is responsible for a specific verification task. The top_tb module acts as top level integration for the complete verification setup. It instantiates Design Under Test (DUT), driver module and scoreboard module. Parameterization is maintained in the top module using WIDTH and RES_WIDTH parameters so that the same testbench can support different operand width without modifying the structure.

The driver module is responsible for generating and applying stimulus to the DUT. It drives all ALU input signals such as OPA, OPB, CMD, MODE, CIN, CE and INP_VALID. A task is implemented for testcase generation and can be easily reused. Inputs are driven at the negative edge of the clock to ensure stable setup timing before the next positive edge, where the DUT samples the inputs.

The scoreboard module acts as the reference model and checker for the ALU outputs. It independently calculates the expected output values based on the applied inputs and compares them with the actual DUT outputs. Expected outputs are generated according to the ALU specification. For multiplication operations, the scoreboard incorporate counter logic for 3 clock cycle delay. Since multiplication requires multiple clock cycles for completion, the scoreboard delays result checking accordingly to match the DUT latency. Intermediate states are tracked using internal counter variable such as count and multi to detect if it is multiplication operation.

The scoreboard performs output comparison during the negative edge of the clock after some delay, ensuring the DUT outputs are stable before checking. If the actual DUT outputs match the expected values, PASS messages are displayed. Otherwise, FAIL messages are displayed.

Timing Behaviour of Testbench:

The timing behaviour of the testbench is designed to accurately synchronize with the operational latency of the ALU DUT. Proper synchronization between the driver, DUT, and scoreboard is essential to ensure correct verification of both single cycle and multiple clock cycle operations.

The driver applies input stimulus at the negative edge of the clock. This ensures that all input signals become stable before the next positive edge of the clock, where DUT samples the inputs.

For normal arithmetic and logical operations, the ALU follows a one cycle latency model. Inputs are applied during one clock cycle and the corresponding outputs are generated during next clock cycle.

Quality of Code Assessment:

The RTL and verification code quality are evaluated using multiple verification and analysis techniques. One of the primary quality assessments is lint checking. Lint analysis helps identify coding issues such as unused signals, incomplete case statements, unintended latch inference, and non-synthesizable constructs. Performing lint analysis improves RTL coding quality and reduces the possibility of simulation and synthesis mismatches.

Code coverage analysis is another important metric used to assess verification completeness. The testcases provided in the driver module aim to cover all arithmetic and logical commands. Functional coverage is improved by applying both normal and corner case stimulus values such as zero, maximum values, signed negative numbers, overflow condition and invalid rotation inputs.

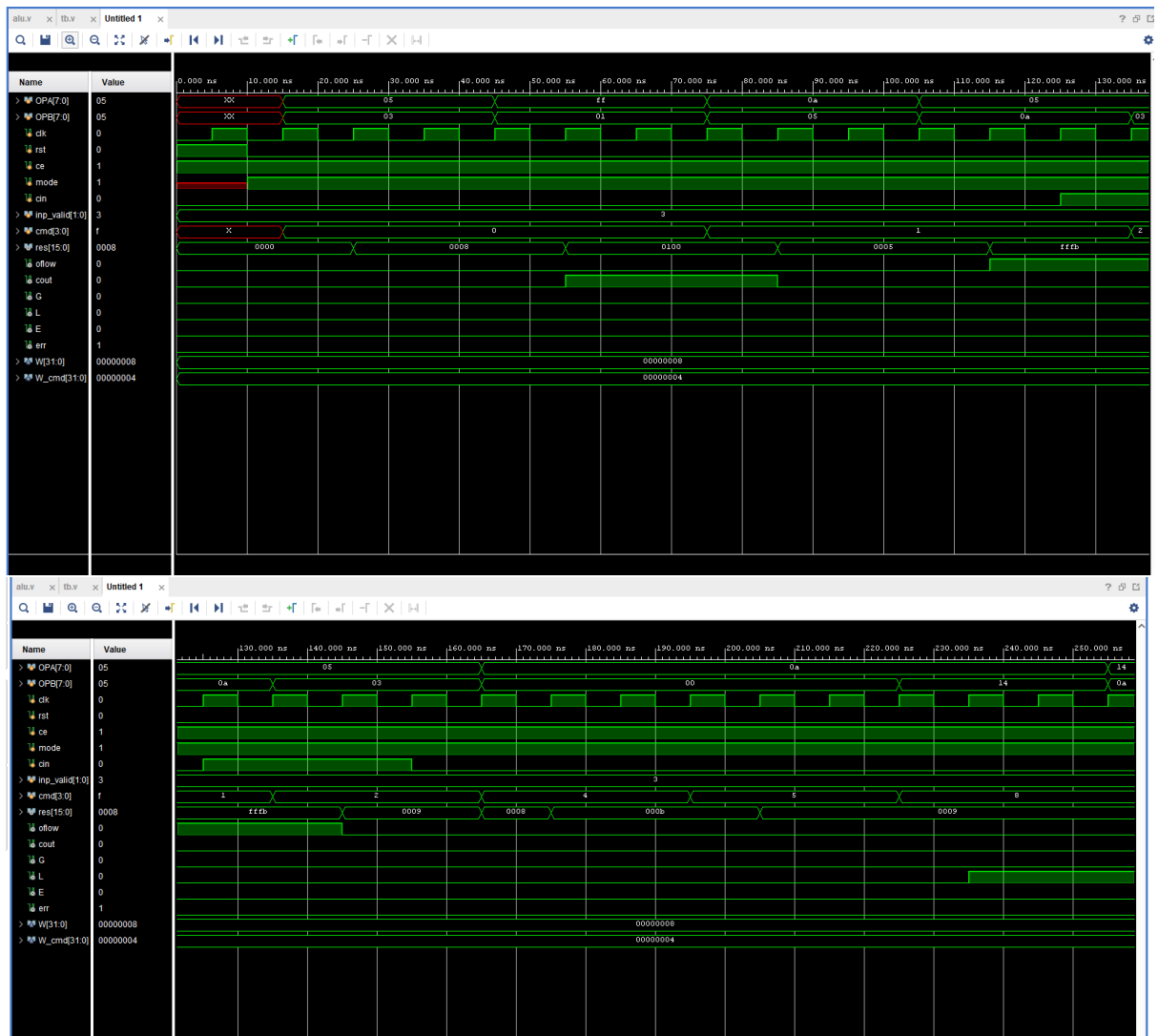
Overall, the verification environment ensures comprehensive validation of the ALU functionality.

- **Results:**

The simulation results demonstrate the functional verification of the parameterized ALU design for both arithmetic and logical operating modes. The design was simulated using a self-checking testbench environment. The arithmetic operation simulations confirmed correct functionality. Similarly, logical operations produced the expected outputs during simulation. Several corner cases were included in the verification process such as zero values, maximum unsigned values, signed negative numbers, overflow conditions. Most arithmetic and logical operations successfully passed the verification checks generated by the scoreboard. However, some multiplication related testcases showed mismatched during simulation. These failures were primarily caused by timing synchronization issues associated with the multi cycle latency. Since multiplication requires additional processing cycles compared to normal single cycle operations, some expected results were compared before the DUT output becomes fully stable.

The verification status of the project indicates that the overall ALU functionality is operational and most features are successfully validated. The project was developed and simulated using Vivado for RTL coding and simulation and waveform-based verification were performed using Questa SIM.

- **Waveform:**





- Coverage Report

← → ↻ file:///home/krishnagirish/Desktop/test/covReport/pages/_frametop.htm

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Testplan Design DesUnits

top_tb

Questa Coverage Report

Number of tests run: 1

Passed: 0

Warning: 1

Error: 0

Fatal: 0

[List of tests included in report...](#)

[List of global attributes included in report...](#)

[List of Design Units included in report...](#)

Coverage Summary by Structure:

Design Scope	Hits %	Coverage %
top_tb	92.34%	96.06%
dut	92.34%	96.06%

Coverage Summary by Type:

Coverage Type	Bins	Hits	Misses	Weight	% Hit	Coverage
Total Coverage: 92.34% 96.06%						
Statements	117	117	0	1	100.00%	100.00%
Branches	104	103	1	1	99.03%	99.03%
FEC Expressions	6	6	0	1	100.00%	100.00%
Toggles	230	196	34	1	85.21%	85.21%

Report generated by Questa (ver. 10.6c) on Sun 10 May 2026 04:44:23 PM IST with command line:
vcover report -html cov1.ucdb -htmldir covReport -details

← → ↻ file:///home/krishnagirish/Desktop/test/covReport/pages/_frametop.htm

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top_tb

Questa Toggle Coverage Report

Show All Show Covered Show Missing

Signal / Value	Hits						ExtMode	Status
	0L->1H	1H->0L	0L->Z	Z->1H	1H->Z	Z->0L		
COUT	0	0	--	--	--	--	--	0.00%
OPA_1[7-0]	0	0	--	--	--	--	--	0.00%
OPB_1[7-0]	0	0	--	--	--	--	--	0.00%

The functional and code coverage analysis for the parameterized ALU was performed using Quest SIM.

The overall DUT coverage achieved during simulation is approximately 96.06% while the total hit percentage is 92.34%. The coverage report indicates that most functional paths and conditions in the ALU design were successfully exercised by the verification environment.

The statement coverage achieved is 100%, indicating that all statement within the DUT were triggered at least once during the simulation.

The branch coverage achieved is 99.03%. The remaining uncovered branch corresponds to an unattainable condition associated with the multiplication operation timing behaviour.

Expression coverage for the DUT is 100% indicating that all logical expressions within the design were verified successfully.

The toggle coverage achieved is 85.21 %. The lower toggle percentage is primarily caused due to two signals which are never driven in the design and the logic for COUT is assigned as zero.

- **Coverage Observation:**

- All major functionalities were successfully covered during simulation
- One branch related to multiplication timing control remained unreachable due to operational constraints.
- Some toggle conditions were not activated because certain internal state is unused.
- COUT logic path remains fixed at logic 0.

- **Conclusion:**

This project successfully implemented and verified a parameterized ALU. The ALU was designed to support both arithmetic and logical operations. The implementation supports multiple functionalities including signed and unsigned arithmetic operations, logical operations, comparison operations, shift and rotate functions, overflow detection, carry generation, error handling, and multi cycle multiplication operations.

A complete self-checking verification environment was developed consisting of driver, scoreboard, and top level testbench architecture. Directed testcases were created to validate normal functionality, corner cases, invalid input conditions and timing synchronization.

- **Future Work:**

Several improvements can be implemented in the future to further enhance the ALU design and verification environment. The current

design can also be extended to support additional arithmetic and logical operation such as bit manipulation.
Additional synthesis optimization and FPGA implementation can also be performed for timing performance.